Open-Source Software

(Redirected from The History and Development of Open-Source Software)

Open-source software is computer software that is distributed along with its source code to the public. This allows users to inspect, modify, and work on the underlying program. This practice developed in the early history of computing where software was shared among researchers and computer manufacturers.[1] When computers became commercialized, companies began to treat software as private property, in which source code was protected. The free software movement, the GNU project, and the development of Linux formalized this method of sharing code. In the late 1990s, this culture had matured into two ideas. The first being “free software”, which took on a philosophical approach based on user freedom, and the second being “open source”, a more business-friendly framework built around collaborative development.[2] Together, these movements have shaped much of the technology used in the industry today.

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The Foundation [edit | edit source]

During the 1960s and 1970s, software was often shared among universities, researchers, and computer users. Its source code was treated as a necessity rather than an ideology of property. Companies and research institutions shared code with the public for debugging, customization, and often academic needs. The Unix operating system was developed by Ken Thompson and Dennis Ritchie in 1969, at Bell Labs. Unix was distributed to universities primarily for academic purposes, and its source code became a tool which helped train a generation of programmers.[1] As the computer industry began to shift towards a commercialized market in the late 1970s and 1980s, companies such as AT&T started restricting access to source code through licensing agreements. This meant that companies started treating software as products to be sold as opposed to knowledge to be shared.[1] This shift from informal collaboration to restricted control from companies set the stage for the pushback in the following decade.

The Free Software Movement [edit | edit source]

In 1983, Richard Stallman, a programmer at Massachusetts Institute of Technology’s Artificial Intelligence Laboratory, announced the GNU Project. The project’s goal was to create a shareable operating system compatible with Unix, so that programmers may never be locked out of software they depended on. Stallman argued that computer users should have the freedom to study, modify, and share software. He believed this issue was an ethical matter rather than a purely technical one.[3]In 1985 he founded the Free Software Foundation, where users were able to freely share software. In 1989, the GNU General Public License (GPL) was developed to give these ideas legal force. The GPL used a “copyleft” structure, which meant that anyone who shared modified versions of GPL-licensed software was required to keep that code free and open.[3][1] This was responsible for turning Stallman’s philosophy into a legal framework, which ensured that no company or individual could close off any shared code.

Linux [edit | edit source]

In 1991, Linus Torvalds, a student at the University of Helsinki, released the source code for a kernel similar to Unix called Linux.[1] On its own, a kernel is only the core of an operating system but, when combined with tools produced by the GNU project, Linux became a complete operating system.[3] The significance of Linux was how it was built. Rather than being developed by a single company or small team, it was developed through contributions of volunteer programmers over the internet. This development rooted in loose coordination proved that large, reliable software could be built without corporate control.[4] It became a key example of what open collaboration was capable of creating.

Open Source [edit | edit source]

By the mid 1990s, Linux’s success had shown that this collaborative model was capable of producing software that was not just ideologically appealing but also technically competitive. In January of 1998, Netscape Communications Corporation decided to release source code for its browser, a major and unexpected move from a commercial software company. A primary driver of this decision was the essay by Eric S. Raymond, “The Cathedral & The Bazaar”.[2] His essay contrasted two development styles, the “Cathedral” model, which examined tightly planned and controlled software projects, and the “Bazaar” model which focused on openly developed code such as Linux. Raymond’s essay argued that the bazaar model could produce better software faster due to the practical advantage of openness.[4] This was when a distinction between “open source” and “free software” began to develop. Stallman’s movement emphasized user freedom and ethical obligation whereas Raymond emphasized efficiency, quality, and business benefits. The following month, Raymond and other programmers met in Palo Alto, California, where the term “open source” was developed to make this framing more appealing to business. This group founded the Open Source Initiative (OSI), which published the official definition of open source to create licensing standards to attract companies to adopt the practice.[2]

Modernization [edit | edit source]

In the late 1990s and early 2000s, companies began to adopt Linux servers. The industry began recognizing the cost savings and flexibility that open-source software offered as opposed to commercial software.[1] After the founding of OSI, open-source licensing was adopted for a number of influential projects. Some notable projects include MySQL database, which became a standard tool for web applications, and the Mozilla Firefox browser, which grew due to Netscape’s decision to open its source code.[5] These examples showed that open-source projects could match commercial software in quality as well as build large developer communities around them. From

this foundation, open-source became fundamental to sectors used throughout the technology industry today, such as cloud computing, cybersecurity, and artificial intelligence. Platforms such as GitHub further transformed this culture by making it easier for programmers anywhere to contribute to a project and collaborate at a scale that would have been impossible during Unix's era of informal sharing.

Conclusion [edit | edit source]

Open-source began as an informal method of collaboration for program development born out of necessity in early computing, and has since developed into an influential culture of shared programming. The free software and GNU movements established the ethical and legal principles for software freedom, while Linux and the Open Source Initiative demonstrated how collaborative software could be adopted by the industry. Though "free software" and "open source" came from the same roots and are often used in similar contexts, they represent two related philosophies. One grounded in user freedom and the other in collaborative efficiency. Together they reshaped how software is shared and built today.

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